Visit fakharstemsphere.com to download it and Model Paper of Math, Class 9th, 2026

MATH Guess Paper Class 9th 2026 Punjab Boards

Objective Type

Q.1

i)

$2^x \times 8^x = 64$ then $x =$

(a) $\frac{3}{2}$ (b) $\frac{3}{4}$ (c) $\frac{5}{6}$ (d) $\frac{2}{3}$

ii)

(v) $\log 100 =$ _____.

(a) 2 (b) 3 (c) 10 (d) 1

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

iii)

If A and B are overlapping sets, then $n(A - B)$ is equal to

(a) $n(A)$ (b) $n(B)$ (c) $A \cap B$ (d) $n(A) - n(A \cap B)$

iv)

The factors of $4x^2 - 12x + 9$ are:

(a) $(2x + 3)^2$ (b) $(2x - 3)^2$

(c) $(2x - 3)(2x + 3)$ (d) $(2 + 3x)(2 - 3x)^2$

v)

If $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $U - (A \cap B)$ is:

(a) $\{1, 2, 4, 5\}$ (b) $\{2, 3\}$ (c) $\{1, 3, 4, 5\}$ (d) $\{1, 2, 3\}$

vi)

The HCF of a^3b^3 and ab^2 is:

(a) a^3b^3 (b) ab^2 (c) a^4b^5 (d) a^2b

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

vii)

If $\sin\theta = \frac{3}{5}$ and θ is an acute angle, $\cos^2\theta =$ _____.

(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) $\frac{16}{25}$ (d) $\frac{4}{25}$

viii)

(0,0) is solution of inequality:

(a) $4x + 5y > 8$ (b) $3x + y > 6$

(c) $-2x + 3y < 0$ (d) $x + y > 4$

ix)

$\cos^2 100\pi + \sin^2 100\pi =$ _____.

(a) 1 (b) 2 (c) 3 (d) 4

x)

A line passing through points (1, 2) and (4, 5) is:

(a) $y = x + 1$ (b) $y = 2x + 3$

(c) $y = 3x - 2$ (d) $y = x + 2$

Visit fakharstemsphere.com to download it and Model Paper of Math, Class 9th, 2026

xi)

If two polygons are similar, then:

(a) their corresponding angles are equal.

(b) their areas are equal.

(c) their volumes are equal.

(d) their corresponding sides are equal.

xii)

The y- intercepts of $y = -2x - 1$ is:

(a) -2 (b) 2

(c) -1 (d) 1

xiii)

Locus of points equidistant from two fixed points is ----------

(a) circle (b) perpendicular bisector

(c) angle bisector (d) parallel lines

Visit fakharstemsphere.com to download it and Model Paper of Math, Class 9th, 2026

xiv)

Midpoint is also known as:

(a) mean (b) median

(c) class limit (d) class mark

xv)

Each element of the sample space is called:

(a) event (b) experiment

(c) sample point (d) outcomes

Subjective Type (Part-I & II)

Express the following as a rational number $\frac{p}{q}$ where p and q are integers and $q \neq 0$:

$$0.\overline{37}$$

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Example 7: Simplify the following:

(i) $\sqrt[4]{16x^4y^8}$ (ii) $\sqrt[3]{27x^6y^9z^3}$ (iii) $(64)^{-\frac{4}{3}}$

Find two rational numbers between:

(i) $\frac{1}{3}$ and $\frac{1}{4}$ (ii) 3 and 4 (iii) $\frac{3}{5}$ and $\frac{4}{5}$

Simplify the following:

$$\frac{54\times\sqrt[3]{(27)^{2x}}}{9^{x+1}+216(3^{2x-1})}$$

(v) $$\frac{5\cdot(25)^{n+1}-25\cdot(5)^{2n}}{5\cdot(5)^{2n+3}-(25)^{n+1}}$$

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

$$\frac{(16)^{x+1}+20(4^{2x})}{2^{x-3}\times 8^{x+2}}$$

$$\frac{(25)^{\frac{3}{2}}\times(243)^{\frac{3}{5}}}{(16)^{\frac{5}{4}}\times(8)^{\frac{4}{3}}}$$

$$\frac{5^{n+3}-6.5^{n+1}}{9\times 5^{n}-4\times 5^{n}}$$

$$\sqrt{\frac{(216)^{\frac{-}{3}}\times(25)^{\frac{1}{2}}}{(0.04)^{\frac{-3}{2}}}}$$

Fakhar Stem Sphere

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Rationalize the denominator of:

(i) $\dfrac{3}{\sqrt{5}+\sqrt{2}}$ (ii) $\dfrac{3}{\sqrt{5}-\sqrt{3}}$

If $x = 3 + \sqrt{8}$ then find the value of:

$x + \dfrac{1}{x}$ (ii) $x - \dfrac{1}{x}$ (iii) $x^2 + \dfrac{1}{x^2}$

Find the value of x in each of the following:

$$\log_4 x = \frac{3}{2}$$

Find logarithm of the following numbers:

0.000354

$$\log x = -1.4567$$

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

What is the number of elements of the power set of each of the following sets?

(iii) $\{1, 2, 3, 4, 5, 6, 7\}$

(v) $\{a, \{b, c\}\}$

$\{\{a, b\}, \{b, c\}, \{d, e\}\}$

Write down the power set of

$\{9, 11\}$

$\{\emptyset\}$ (iv) $\{a, \{b, c\}\}$

Visit fakharstemsphere.com to download it and Model Paper of Math, Class 9th, 2026

Define the Union, Intersection, disjoint set, overlapping sets, and complement of a Set.

In a class of 55 students, 34 like to play cricket and 30 like to play hockey. Also each student likes to play at least one of the two games. How many students like to play both games?

Factorize: $x^2 + 9x + 14$

Factorize: $x^2 - 11x + 24$

Factorize: $2x^2 + 17x + 26$

Factorize:

$x^6 - 27$

$27 - 512y^3$

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

$1000a^3 + 1$ (v) $343x^3 + 216$

$8x^3 + 12x^2 + 6x + 1$ $x^4 - 14x^2 + 1$

$a^4 + 64b^4$ (iii) $x^4 + 4x^2 + 16$

Factorize: $(x + 2)(x + 3)(x + 4)(x + 5) - 15$

Factorize: $(x - 2)(x + 2)(x + 1)(x - 4) + 2x^2$

Factorize: $(x^2 + 5x + 4)(x^2 + 5x + 6) - 3$

$x^4 - 14x^2 + 1$

$x^4 - 7x^2y^2 + y^4$

(iv) $8x^3 - 125y^3 + 150xy^2 - 60x^2y$

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Solve the following equations and represent their solutions on real line:

$$3x-5=7$$

Find solution of $\frac{2}{3}x-1<0$ and also represent it on a real line.

Define Coterminal Angles.

Example 1: Convert 73.12° to degrees, minutes, and seconds.

Example 2: Convert 109.42° to degrees, minutes, and seconds.

Example 5: Convert radians to degrees

(i) $\frac{5\pi}{3}$ rad (ii) $\frac{7\pi}{6}$ rad (iii) $\frac{11\pi}{6}$ rad (iv) 1.2 rad

Example 6: Convert degrees to radians

(i) 15° (ii) 75° (iii) 315° (iv) 15° 15′

Example 7: Find the arc length of a sector with radius r = 10 cm and central angle θ = 60°.

Find the area of a sector with radius r = 8 cm and central angle θ = 45°.

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Evaluate:

(i) $2 \sin 60° \cos 60°$

(iii) $2 \sin 45° + 2\cos 45°$

Example 14: If $\tan \theta = \frac{3}{4}$, find the remaining trigonometric ratios, when θ lies in first quadrant.

Example 9: If arc length of a sector of radius 5 cm is 11 cm, find the angle subtended by the arc in radians and degrees.

(v) $\cos 60° \cos 30° - \sin 60° \sin 30°$ (vi) $\sin 60° \cos 30° - \cos 60° \sin 30°$

Show that $\tan \theta + \cos \theta = \sec \theta \ \operatorname{cosec} \theta$

Example 15: Solve triangle ABC, in which $m\angle B = 90°$, $m\angle A = 30°$, $a = 2$ cm

Show that $(\sec^2 \theta - 1) \cos^2 \theta = \sin^2 \theta$

Visit fakharstemsphere.com to download it and Model Paper of Math, Class 9th, 2026

$\cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$

7. $\dfrac{1-\sin\theta}{\cos\theta} = \dfrac{\cos\theta}{1+\sin\theta}$

$(\sec\theta - \tan\theta)^2 = \dfrac{1-\sin\theta}{1+\sin\theta}$

9. $(\tan\theta + \cot\theta)^2 = \sec^2\theta\,\text{cosec}^2\theta$

$$\frac{\sin\theta}{\text{cosec}\,\theta} + \frac{\cos\theta}{\sec\theta} = 1$$

Find h such that $A(-1, h)$, $B(3, 2)$ and $C(7, 3)$ are collinear.

The points $A(-5, -2)$ and $B(5, -4)$ are ends of a diameter of a circle. Find the centre and radius of the circle.

the points $A(3, 1)$, $B(-2, -3)$ and $C(2, 2)$ are vertices of an isosceles triangle.

the points $A(5, 2)$, $B(-2, 3)$, $C(-3, -4)$ and $D(4, -5)$ are vertices of a parallelogram.

Find k so that the line joining $A(7, 3)$; $B(k, -6)$ and the line joining $C(-4, 5)$; $D(-6, 4)$ are:

(i) parallel (ii) perpendicular.

Using slopes, show that the triangle with its vertices $A(6, 1)$, $B(2, 7)$ and $C(-6, -7)$ is a right triangle.

Visit fakharstemsphere.com to download it and Model Paper of Math, Class 9th, 2026

Define Distance Formula between Two Points

Example 1: Find the distance between the points:

(i) $A(5, 6), B(5, -2)$ (ii) $C(-4, -2), D(0, 9)$

Find the midpoint of the line segment joining the points A (2,3) and B(8,7).

Define Slope or Gradient of a line.

Find an equation of the straight line if

(a) its slope is 2 and y-intercept is 5

Find the slope and inclination of the line joining the points:

(i) (–2, 4) ; (5, 11) (ii) (3, –2) ; (2, 7) (iii) (4, 6) ; (4, 8)

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Find an equation of:

(b) the vertical line through $(-5, 3)$

through $A(-6, 5)$ having slope 7

through $(-5, -3)$ and $(9, -1)$

y-intercept: -7 and slope: -5

Find an equation of the line through $(-4, -6)$ and perpendicular to a line having slope $\frac{-3}{2}$.

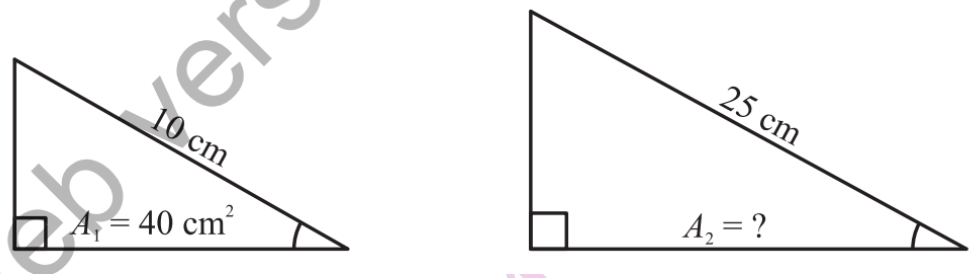
Find an equation of the line through $(11, -5)$ and parallel to a line with slope -24.

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Example 6: Find the unknown value in the following:

(i)

(ii)

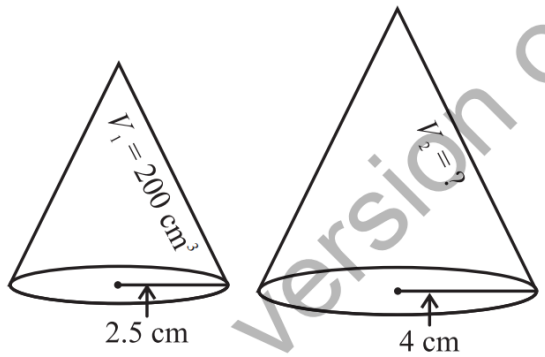


Web version of PCTB
Fakhar STEM

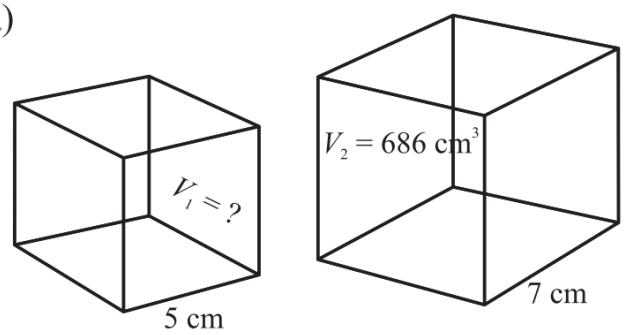
Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Example 9: Find the unknown volume in the following similar solids:

(i)



(ii)



2. Two regular tetrahedrons have volumes in the ratio 8 : 27. What is the ratio of their sides?

3. Two right cones have volumes in the ratio 64 : 125. What is the ratio of:
   (i) their heights (ii) their base areas?

Sketch the graph of the following linear functions:

(i) $y = 3x - 5$ (ii) $y = -2x + 8$

(iii) $y = 0.5x - 1$

Plot the graph of the following functions:

$$y = 4^x \qquad y = \frac{1}{x-3} \quad x \neq 3 \qquad y = x^{-1}$$

Visit fakharstemsphere.com to download it and Model Paper of Math, Class 9th, 2026

Define Scalene, Isosceles, Right and Equilateral Triangle.

Define Circumcenter, Incentre and Centroid.

Define Median and Mode of Ungrouped Data.

Find the arithmetic mean in each of the following:

(i) 4, 6, 10, 12, 15, 20, 25, 28, 30.

(ii) 12, 18 19, 0, –19, –18, –12

(iv) 8, 10, 12, 14, 16, 20, 22

Following are the earnings (in Rs.) of ten workers:

88, 70, 72, 125, 115, 95, 81, 90, 95, 90. Calculate

(i) Arithmetic Mean (ii) Median (iii) Mode

The marks obtained by the students of Grade - 10 in mathematics test were grouped into the following frequency distribution.

Marks	35 – 37	38 – 44	45 – 54	55 – 61	62 – 67	68 – 72
Frequency	2	12	16	13	9	3

Draw a histogram for the above distribution.

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Example 6: Given below are the marks out of 100 obtained by 100 students in a examination. Find the average marks of the students.

Marks	30 – 35	35 – 40	40 – 45	45 – 50	50 – 55	55 – 60
No. of students	14	16	18	23	18	11

Example 7: Find the arithmetic mean using short formula for the runs made by a batsman.

Runs: 40, 45, 50, 52, 50, 60, 56, 70.

Define Event , Outcomes and Favorable Outcome.

Arshad rolls a dice, with sides labelled L, M, N, O, P, U. What is the probability that the dice lands on consonant?

Aslam rolled a dice. What is the probability of getting the numbers 3 or 4? Also find the probability of not getting the numbers 3 or 4.

Visit fakharstemsphere.com to download it and
Model Paper of Math, Class 9th, 2026

Example 8: Find the relative frequency of the given date.

X	2	3	4	5	6	7	8
f	3	5	6	9	10	8	2

Example 10: Abdul Rehman obtained different marks in different subjects out of 100 marks. The detail is as under:

Subject	Urdu	English	Islamiyat	Mathematics	Science	Computer Science
Marks Obtained	75	80	72	95	81	85

Find the relative frequency of above given data.

Fakhar STEM
PCTB textbook